A position-only Checkerboard shell element for target-shape design

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Abstract

This draft contains the Introduction only.

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1. Introduction

1.1. A Position-Only Shell from a Checkerboard Pattern

Thin shells carry loads efficiently by combining membrane action with bending, but their design and active deformation are complicated by the rotational degrees of freedom carried by most shell elements. A position-only element would make the geometry both the analysis state and the design variable; what is missing is a discrete surface that supports a complete shell energy with positions alone. This paper uses a checkerboard pattern for that purpose.

Consider a quadrilateral grid whose vertices are colored in a checkerboard pattern. Discard the vertices of one color and keep the midpoints of the edges of the remaining quads. Each quad is then replaced by the four midpoints of its edges. These four points are coplanar and form a parallelogram, the

Varignon parallelogram, for any spatial position of the original quad. The resulting surface is a collection of flat quadrilateral facets whose vertices are edge midpoints of the original grid.

The construction is attractive for a shell element. The local tangent plane and normal follow directly from nodal positions; no diagonal choice is required; and the membrane metric and the bending curvature can share one geometric carrier. The element developed here uses one position vector per node, three degrees of freedom per node, and no rotational degrees of freedom.[1] The question is whether this minimal state can carry a complete shell energy, and whether the resulting element is useful when the shell geometry itself is the design variable.

1.2. Checkerboard Patterns in Geometry Processing

Checkerboard and midpoint constructions have a long history in geometry processing. In discrete differential geometry they appear as discrete isothermic surfaces and checkerboard nets [1, 2]. In graphics, position-based and midedge shell models use edge midpoints to define bending [3, 4], while discrete shape operators and subdivision shells build curvature from positions [5, 6, 7]. Smooth-basis isogeometric shells take a different route: they carry exact geometry through the basis [8, 9]. The appeal of the midpoint pattern is its locality, exact planarity, and the fact that one carrier serves both membrane and bending. The pattern is attractive not only because it looks right: the local frames and fundamental forms are available in closed

[1] Here "Checkerboard" refers to the midpoint pattern, not the checkerboard instability of mixed finite elements.

form, which makes it a convenient primitive for geometry-driven design.

Here we use the geometric primitive as the starting point for a mechanical formulation. The question is mechanical: whether the midpoint surface can carry a reliable shell energy under large deformation, and whether it can support design loops in which geometry is the variable.

1.3. Why Bring It into Mechanics

The motivation goes beyond turning a geometric pattern into a shell. A position-only element would change how shell design is posed. First, the state is the design variable: nodal positions are both the analysis unknown and the geometric design variable; no rotational parameterization or rotational constraint is needed, and the adjoint unknown has the same three-degree-of-freedom structure as the state. Second, design loops become cheaper: target-shape design, actuator layout, and shape optimization require one forward and one adjoint solve per gradient instead of $2N$ forward solves. Third, large rotations are handled by position updates: deployable and folding structures do not need a rotation parameterization in the design loop. Fourth, one geometric carrier serves membrane and bending: the same midpoint surface provides both the metric and the curvature, without a separate director field.

These benefits point to applications such as morphing skins, active panels, deployable booms and stents, soft actuators, and compliant mechanisms. They are, however, potential benefits. A geometric pattern that looks natural can still produce a discrete energy with spurious soft modes or a wrong thickness scaling. Whether the benefits are realized depends on a mechanical question, which we turn to next.

1.4. From Discrete Geometry to Mechanics: The Gap

Geometry processing asks whether a discrete surface approximates a smooth one. Mechanics asks a different question: whether the discrete energy is complete at the lattice scale and in the thin-shell limit. Three issues matter. First, the metric and the symmetric curvature are local averages; they can be consistent to second order on smooth data and still be blind to lattice-scale modes. Second, membrane and bending energies scale with thickness as t and t^3 [10, 11]; a completion that does not respect this separation can turn a bending mode into a membrane mode. Third, the energy must remain objective under rigid rotation, especially when the design loop explores large rotations [12].

We call the modes that the midpoint metric and symmetric curvature cannot see *blind modes*. For the periodic midpoint surface, the analysis identifies one membrane mode and two curvature modes. The migration from discrete geometry to mechanics is therefore a formulation problem: identify these modes and complete them without destroying the bending space. The central question is whether a minimal, targeted completion can do that while preserving the simplicity of the position-only state. This is the same tension that appears in membrane locking, where consistency of the strain approximation is not enough and the discrete kernel matters [13, 14, 15].

1.5. This Paper's Approach

The paper follows four steps. We first define the midpoint surface, the flat B faces that carry the membrane metric, and the W-centered construction that carries the second fundamental form. The same midpoint surface serves both; boundary closure is local. We then analyze what the midpoint metric

and symmetric curvature miss. The membrane metric misses one alternating in-plane mode; the symmetric curvature misses two transverse modes. We construct one scalar completion for the membrane mode and one objective, reference-relative curvature completion for the two curvature modes. Each completion acts only on the mode it is designed for.

The nonlinear tangent inherits this structure. It is a sparse local matrix plus one global rank-one term; the global term is solved with a bordered system and is never assembled as a dense outer product. This makes the tangent exact for the discrete energy and the design loop cheap. Finally, we use the element for target-shape design: given a desired shape, solve for actuator forces or for a low-dimensional set of initial geometry parameters. The examples are chosen to expose both the capability and the boundary of the inverse problem.

1.6. Contributions

The contributions are fourfold. First, we construct a three-degree-of-freedom-per-node shell element from the checkerboard midpoint geometry; the state is nodal position only, and the membrane and bending energies share one geometric carrier. Second, we identify the one membrane and two curvature modes that the midpoint metric and symmetric curvature miss, and give two targeted completions that act only on those modes. Third, we derive an exact nonlinear tangent that is local and sparse apart from one global rank-one term, and solve the global term with a bordered system. Fourth, we demonstrate target-shape design: a nine-actuator shape-control problem reaches a 0.0065% target error, a two-parameter initial-curvature design through a 90° rotation recovers the target to 10^{-6}, and the adjoint

design loop needs about 5.7 times fewer linear solves than finite differences on the same model.

1.7. Scope and Preview

The element is aimed at design, active deformation, and large rotation. We calibrate it on standard shell benchmarks and compare it with a mid-edge shell [4] and MITC4 [16, 17]. The comparison establishes where the element sits, with the benchmarks used for calibration. The operating range, including the mesh-thickness rule, is stated in Section 4; the trade-offs are discussed in Section 7.

The remainder is organized as follows. Section 2 defines the element. Section 3 identifies the blind modes and constructs the completions. Section 4 presents analytical properties and numerical implementation. Section 5 reports verification and benchmarks. Section 6 presents the design applications. Section 7 discusses the scope and concludes.

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